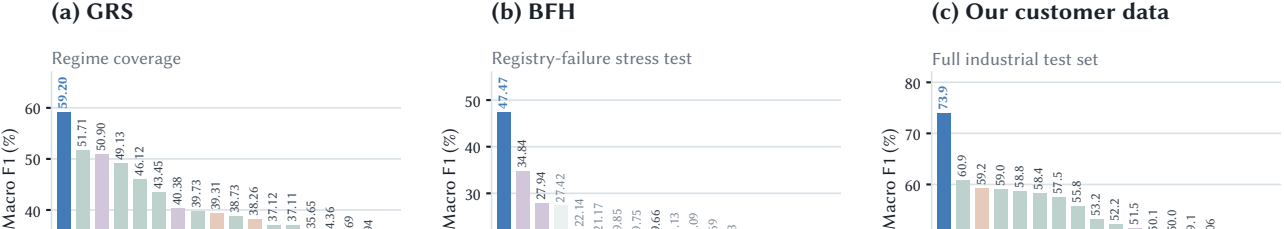
(a) GRS
Regime coverage
Macro F1 (%)
59.20
51.71
50.90
49.13
46.12
43.45
40.38
39.73
39.31
38.73
38.26
37.12
37.11
35.65
34.36
32.69
31.94
29.20
Agentic4PDM
M1.1R-SVM
ZARA
M1.1R-XGB
M1.1R-KNN
C2-SVM
FETA
M2S-SVM
QCNN
M2S-XGB
WDCNN
C2-KNN
C2-XGB
C2S-SVM
C2S-XGB
M2S-KNN
CNN-TCC
C2S-KNN
(b) BFH
Registry-failure stress test
Macro F1 (%)
47.47
34.84
27.94
27.42
22.14
21.17
19.85
19.75
19.66
19.13
19.09
17.69
16.43
14.63
14.29
12.84
12.53
10.54
Agentic4PDM
ZARA
FETA
M1.1R-XGB
M2S-KNN
M2S-XGB
M1.1R-SVM
M1.1R-KNN
CNN-TCC
C2S-KNN
C2-SVM
C2-KNN
M2S-SVM
C2S-XGB
C2-XGB
WDCNN
QCNN
C2S-SVM
(c) Our customer data
Full industrial test set
Macro F1 (%)
73.9
60.9
59.2
59.0
58.8
58.4
57.5
55.8
53.2
52.2
51.5
50.1
50.0
49.1
47.06
44.2
44.1
42.9
Agentic4PDM
C2S-SVM
WDCNN
C2-KNN
M2S-SVM
C2S-XGB
M1.1R-SVM
M2S-XGB
C2-XGB
C2-SVM
FETA
M1.1R-XGB
C2S-KNN
QCNN
M1.1R-KNN
ZARA
CNN-TCC
M2S-KNN
Agentic4PDM
Agent baselines
Foundation models
Deep learning